

Rishabh Bhattarai

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EDUCATION

University of Michigan - College of Engineering | Ann Arbor, MI

April 2027

B.Sc. Electrical and Computer Engineering | GPA: 3.7

Coursework: VLSI Design, Digital Circuits, Computer Architecture, Signals and Systems, Data Structures & Algorithms

WORK EXPERIENCE

FPGA Engineer Co-op | Sierra Nevada Corporation | Folsom, CA

January 2026 - Present

- Develop synthesizable VHDL RTL for FPGA-based DSP and radio systems in Vivado, designing custom datapath modules, integrating vendor IP cores, and resolving timing, pipelining, and synchronization challenges.
- Design and maintain VHDL testbenches to verify RTL functionality across edge cases and corner conditions, performing simulation, waveform debugging, and iterative validation to identify issues and improve designs.
- Implement new FPGA features by modifying RTL, AXI-based interfaces, and register maps, supporting hardware/software interaction through system-level simulation, verification, and timing analysis.
- Develop and maintain Tcl scripts to automate simulation setup, execution, and debug workflows in Vivado, reducing manual setup effort and improving verification efficiency for FPGA development and testing.

Systems Engineer Intern | Sierra Nevada Corporation | Englewood, CO

May 2025 - December 2025

- Developed a reusable SysML-based Model-Based Systems Engineering (MBSE) meta-model used across 4+ engineering teams, improving hardware/software interface traceability and reducing design rework by ~25%.
- Integrated an Arduino-powered robotic arm with SNC internal tools using C++, developing interface software, validating end-to-end connectivity, and documenting integration specifications for future third-party tool support.
- Collaborated with 10+ cross-functional engineers on architecture development, lifecycle modeling, and modernization of legacy processes, contributing to multiple requirement reviews for aerospace and defense systems.

SharePoint Developer | MSU Facility for Rare Isotope Beams | East Lansing, MI

May 2024 - January 2025

- Developed and deployed SharePoint Online applications and Power Automate workflows, reducing manual effort by ~30% while supporting 500+ engineering users through app deployments, troubleshooting, and system maintenance.

PROJECTS

16-Bit RISC Processor with 4x4 Systolic Array Accelerator (Cadence Virtuoso/Innovus, Verilog)

May 2026

- Extended a 2-stage pipelined 16-bit RISC processor with a 4x4 systolic array accelerator for matrix multiplication, integrating custom datapath hardware, SRAM interfaces, and a synthesized controller.
- Designed fully custom VLSI layouts for the register file, ALU, and logarithmic shifter in Cadence Virtuoso; performed schematic capture, transistor-level CMOS design, DRC/LVS verification, synthesis with Synopsys Design Compiler, and place-and-route (P&R) using Cadence Innovus as part of a full VLSI design flow.
- Developed Verilog RTL for processor control and accelerator integration, implementing FSM-based control logic and custom ISA instructions; achieved an 8x speedup in matrix multiplication versus baseline processor execution.
- Integrated a dual-port 16x16 SRAM into a unified processor datapath and analyzed post-P&R timing, area, power, and performance tradeoffs across a complete RTL-to-GDSII ASIC design flow, evaluating physical design impacts.

8-Bit Dual-Mode Ripple Carry Adder (Cadence Virtuoso, Maestro, CMOS/VLSI Design)

December 2025

- Designed and implemented a complete 8-bit dual-mode ripple carry adder at the transistor level, including full adders, registers, multiplexers, and inverter networks in Cadence Virtuoso for arithmetic and logical operations.
- Built a full Spectre-based simulation and verification environment with clocked testbenches, vector-driven inputs, and unified power rails to validate functional correctness, timing, and power constraints across multiple conditions.
- Performed transistor sizing studies to create high-performance and low-power design variants, analyzing delay, dynamic power, and leakage under representative switching activity, while evaluating key design tradeoffs.

TECHNICAL SKILLS

Skills: Verilog, VHDL, Vivado, Cadence Virtuoso/Innovus, Linux, Git, Tcl, Python, C/C++, RTL Verification, SPICE, AXI